

Ziji Sheng

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Education

Huazhong University of Science and Technology

Sep. 2024 – Jun. 2027 (expected)

Master of Engineering; School of Cyber Science and Engineering.
GPA: **3.81/4.00**. National Scholarship (2026).

Wuhan University

Sep. 2020 – Jun. 2024

Bachelor of Engineering; School of Cyber Science and Engineering.
GPA: **3.62/4.00**.

Research Interests

- Safety, trustworthiness, and applications of multimodal large language models and multi-agent systems.
- Multimodal large language models and embodied intelligence.

Publications

* Equal contribution; applicant's name in bold.

1. **Ziji Sheng**^{*}, Guiyao Tie^{*}, Weidong Wang^{*}, Jiawen Shi^{*}, Junhao Dong, Xiaoye Qu, Shanshan Ye, Daizong Liu, Dengpan Ye, Pan Zhou, and Jing Zhang.
Object Hallucination Mitigation in Large Vision-Language Models via Self-Vision Dual Masking and Uncertainty-Triggered Assembly.
NeurIPS 2026, Main Track.
 - SIGMA separates supporting visual evidence from potentially misleading context within the model's own vision encoder, using dual masks and uncertainty-triggered three-stream assembly.
 - Training-free, without external vision experts; on LLaVA-1.5, average POPE F1 improves by 5.51 percentage points over regular decoding, with 0.2152 s/sample versus AGLA's 0.4697 s/sample.
2. **Ziji Sheng**^{*}, Guiyao Tie^{*}, Weidong Wang, Pan Zhou, and Daizong Liu.
LEARNERCOMPASS: Intelligent Tutoring System with Dynamic Cognitive Diagnosis and Multi-Model Path Planning.
ACL 2026, Main Conference (Long Paper), pp. 9016–9042.
 - Integrates dynamic knowledge graphs, Graph-RAG, multi-agent cognitive diagnosis, and adaptive branching Monte Carlo tree search for personalized learning-path planning.
 - Reduces hallucination rate from 31.5% for the base LLM to 4.1% for the full system in the paper's educational-task evaluation.
3. Lin Lu^{*}, Zhigang Zuo^{*}, **Ziji Sheng**^{*}, and Pan Zhou.
Merger-as-a-Stealer: Stealing Targeted PII from Aligned LLMs with Model Merging.
EMNLP 2025, Main Conference, pp. 5806–5825.
 - Identifies a privacy attack surface in model merging: malicious model parameters can compromise alignment and enable targeted extraction of personally identifiable information.
 - Evaluates the attack across LLMs and merging methods; achieves 76% exact-match email extraction against DPO-aligned LLaMA-2 in the reported setting.

Manuscript Under Review

PersistInject: Persistent Prompt Injection Attacks against Multi-Turn LLM Applications.

Co-first author; *IEEE Transactions on Information Forensics and Security*, under review.

Studies persistent prompt injection through malicious instructions embedded in model responses and staged optimization, allowing an initial injection to influence subsequent interactions.

Ongoing Research

Uncertainty-Aware On-Policy Distillation for Reliable VLA Models

Investigating whether hallucinated teacher predictions transfer to student policies and whether grounding errors compound during closed-loop execution. Exploring uncertainty and visual/instruction consistency signals to assess teacher-supervision reliability and guide selective weighting or verification during distillation. *Work in progress.*

Selected Research and Project Experience

Human-AI Collaborative Planning

Nov. 2024 – May 2026

Student Project Lead | Provincial/Ministerial-Level Research Project

- Coordinated task decomposition, technical design, development of three core modules, and integration for collaborative planning under incomplete information and changing constraints.
- Combined human-AI decision making, an adaptive knowledge base, and feedback-driven strategy refinement; delivered a prototype and completed project acceptance.

Multi-Agent Adaptive Learning-Path Planning

May 2025 – Oct. 2025

Team Lead | Challenge Cup National First Prize

- Led system design and algorithm development for knowledge-grounded reasoning and personalized planning, integrating dynamic knowledge graphs, Graph-RAG, and multi-agent diagnosis.
- Combined AB-MCTS-M path search with reflection to adapt learning sequences to evolving cognitive states; this research direction is represented by LEARNERCOMPASS.

Temporal RAG and Multi-Agent Security-Alert Analysis

Jul. 2025 – Oct. 2025

Project Lead

- Designed a layered pipeline for multi-source security-log screening, contextual retrieval, and coordinated analysis; implemented temporal RAG and specialist-agent orchestration.
- Integrated Redis, Milvus, and Neo4j for temporal, vector, and graph-based context storage and retrieval; used LangGraph to orchestrate Qwen-based agent workflows.

Knowledge-Graph Construction on Domestic Computing Platforms

Mar. 2023 – Sept. 2023

Core Contributor | Challenge Cup National Grand Prize and Champion

- Implemented entity/relation extraction with Paddle UIE and iteratively refined annotation schemes; contributed to a knowledge-graph pipeline with reasoning, entity resolution, and Neo4j.

Research-Relevant Industry Experience

ByteDance – Feishu (Lark)

May 2026 – Sept. 2026

LLM / Agent Security Algorithm Intern

- Developed an input-semantic pipeline combining MacBERT intent classification and UIE entity extraction; constructed 10,000+ intent and 21,000+ extraction examples. Reported evaluation: average intent precision/recall above 80%; extraction macro-F1 of 76–78%.
- Designed evidence linking sensitive resources, export actions, and external destinations across user input, tool calls, and API execution; contributed to event and information-flow representations for explainable, replayable agent-security analysis.

Selected Honors and Awards

- National Scholarship**, China, 2026.
- National First Prize**, 19th Challenge Cup, Special Competition, 2025; Team Lead.
- National Grand Prize and Champion**, 18th Challenge Cup, Special Competition, 2023; Core Contributor.
- National Second Prize / Third Prize**, 3rd / 4th China Postgraduate Cybersecurity Innovation Competition, respectively.
- National Third Prize**, 19th National College Student Information Security Competition (Project Track).
- Merit Student and Outstanding Communist Youth League Member, HUST; Outstanding Student Leader (2020–2021), Wuhan University.

Granted Invention Patents

Three Chinese invention patents; **second-listed inventor** on each; first-listed inventor: Pan Zhou. Patent owner: HUST.

- Adaptive Learning-Path Planning Based on Multi-Agent Deep Reasoning**. ZL 2025 1 1052184.0; granted Sep. 22, 2026.
- Network Security-Alert Analysis Using Multi-Agent LLMs and Dynamic RAG**. ZL 2026 1 0679185.6; granted Aug. 14, 2026.
- Multi-Agent Collaborative Retrieval Based on LLM Causal Verification**. ZL 2026 1 0711361.X; granted Aug. 14, 2026.

Technical Expertise

Research methods: LLM safety evaluation, hallucination mitigation, model merging, knowledge graphs, Graph-RAG, multi-agent planning, and uncertainty-guided inference.

Frameworks and systems: PyTorch, Transformers, vLLM, Paddle UIE, LangGraph, Neo4j, Milvus, Redis, and Linux.